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TANVIR AHMED SIJAN

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Research Interests

LLM Interpretablility, Cultural and Low Resource NLP, AI Safety

Education

Jahangirnagar University Dhaka, Bangladesh
Master of Science in Computer Science (PMSCS track) *May 2026 (Expected)*

- **CGPA**: 3.97/4.0
- **Relevant Coursework**:
  * **Machine Learning & AI:** Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Artificial Intelligence, Computational Intelligence
  * **Core Computer Science:** Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Computer Architecture
  * **Systems & Software:** Programming Languages, Software Engineering

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
Bachelor of Science in Civil Engineering, with Honors *Mar. 2025*

- **CGPA**: 3.77/4.0, **Major**: Transportation Engineering, **Minor**: Environmental Engineering
- **Relevant Coursework**:
  * **Mathematics:** Differential and Integral Calculus, Matrices, Laplace Transform and Vector Analysis, Numerical Methods, Applied Mathematics for Engineers
  * **Statistics:** Probability & Statistics
  * **Programming:** Computer Programming Sessional (Structured Programming, OOP), Engineering Computation
  * **Applied:** Transportation Engineering, Structural Analysis, Urban Transportation Planning

Research Experience

Graduate Researcher August 2025 – Present
JU CIDL Research Lab *Supervisor: Dr. Md Musfique Anwar*

- **Event Detection in Bangla Text: Assessing Language Model Robustness to Noisy Data**
  Research Project, Manuscript in preparation
  * Developed a robust event detection framework for Bangla to benchmark the performance of instruction-tuned LLMs against encoder-based models on noisy, ASR-transcribed news text.
  * Synthesized perturbed datasets to quantify model degradation thresholds, specifically analyzing the impact of Word Error Rate (WER) on the stability of event trigger extraction.
- **BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction** Jan – May
  Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar
  Accepted (Oral)** at **ACL Student Research Workshop (SRW) 2026 [arXiv]
  * Conceptualized and led the development of BanglaSocialBench, a benchmark comprising 1,719 culturally grounded instances across three domains to evaluate LLM sociopragmatic competence.
  * Benchmarked 12 LLMs in zero-shot setting, uncovering over-politeness, religious kinship term conflation, and rigidity in ambiguous social contexts.

Undergraduate Thesis Nov. 2023 – Feb. 2025
Bangladesh University of Engineering and Technology (BUET) *Supervisor: Dr. Md. Shamsul Hoque*

- **Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data**
  Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque
  Submitted to Accident Analysis & Prevention, 2025

* Built an end-to-end multimodal data pipeline integrating Google Street View API, Overture Maps, OpenStreetMap, and crash datasets to generate urban risk features for pedestrian crash severity prediction.
* Applied semantic segmentation (Detectron2 + Mask2Former) on 12K+ street-view images and trained ensemble ML models (XGBoost, Random Forest, AdaBoost, CatBoost) with SHAP-based interpretability to identify key environmental risk factors.

Work Experience

Maks Inc. — ***Dhaka, Bangladesh (HQ: Frisco, TX, USA)***
Research Intern — *May 2025 – Sep 2025*

Projects: World Bank Heavy-Duty Vehicles (HDV) Baseline Assessment, Akhaura–Tamabil Land Port Feasibility Study

- Modeled baseline HDV air pollution metrics using the MOVES5 and large-scale BRTA fleet data.
- Used QGIS for spatial analysis and supported field-data collection for infrastructure assessment studies.
- Conducted stakeholder interviews across 7 administrative divisions with transportation operators, associations, and local leaders to document operational and socioeconomic conditions.
- Performed literature reviews and contributed to technical reports with project engineers and research scientists.

Honors & Awards

Undergraduate Student Research Grant — Jan. 2024
Research and Innovation Centre for Science and Engineering, BUET
- Awarded competitive research funding for undergraduate research project on **Deep Learning-based analysis** and **SHAP-based interpretability** of pedestrian crash severity using public geospatial data.

Dean's List Scholarship — Jul. 2019
Bangladesh University of Engineering and Technology (BUET)
- Awarded for maintaining an average GPA of 3.75 or above across two consecutive regular terms.

3rd Place, Concrete Solutions Competition — Mar. 2022
American Concrete Institute (ACI)
- Recognized for the most innovative team design; project featured in the **Daily Prothom Alo** [Feature]

Higher Secondary Certificate (HSC) Scholarship — 2019
Board of Intermediate and Secondary Education, Rajshahi
- Awarded **Talentpool Scholarship**; ranked **10th (top 0.01%)** among all candidates in the Rajshahi Board.

Academic Projects

Mohr's Circle and 2D Stress Calculator App using MATLAB
Engineering Computation Sessional

- Developed an interactive application using MATLAB App Designer that can plot Mohr's Circle from stress block inputs, identify max/min stresses, display plane orientations, and export plots. [GitHub]

Leadership & Voluntary Works

ITE BUET Student Chapter — *Dec. 2024 – Mar. 2025*
Deputy Vice President
- Collaborated with other executive members to organize workshops and seminars on various topics of transportation research.

ACI BUET Student Chapter — *Mar. 2024 – Mar. 2025*
Vice President
- Organized poster competitions, seminars, and field visits in coordination with chapter members.

Skills

Technical Skills
- **Languages:** C, C++, Python, JavaScript, SQL
- **Libraries:** NumPy, PyTorch, Matplotlib, scikit-learn, Selenium
- **Other:** Git, LaTeX, Bash